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The sun is a luminous gaseous sphere with a surface temperature of at least 6000°C . The temperature at the inner core is estimated to be at least 2 million $^{\circ}\text{C}$. The sun emits energy in form of electromagnetic waves, which travel radially

factwards in straight lines at a speed of about 299,300 km/sec solar radiation takes only $9\frac{1}{3}$ min to travel about 150 million km. distance between the sun and the earth.

the X-ray and the earth's.

Although beta radiation travels through space with loss of energy, the intensity of radiation decreases inversely as the square of the distance from the source.

from the Sun. This explains why the further a planet is from the Sun, the colder that planet is. The output of energy from the Sun is

The output of energy from the Sun is
Virtually constant, so also the amount of solar

Virchally Constant, So also the amount of solar energy the earth intercepts. The amount of solar

Energy received per unit area of surface called

at right angle to the rays of the sun on top

⑦ The atmosphere is about $2 \times 10^{-10} \text{ m}^2/\text{min}$ or

2. Kinglets. This is known as the Star constant

Dr. J. W. C. Brown, Jr., 1875-1876

The variation about the average value of $2 \text{ Cal/cm}^2/\text{min}$ being only 2%.

ENERGY BUDGET CONCEPT.

The energy budget concept is a concept used in climatology to specify how the net radiation available at a given surface is used. It is expressed in the equation of the form:

$$R_n = LE + H + G + \Delta F + P$$

where
 R_n = net radiation
 LE = latent heat of evaporation
 H = sensible heat
 G = heat transported into or out of storage
 ΔF = net horizontal advection of heat & T_{surface}
 P = energy used for photosynthesis.

Energy used for photosynthesis is very small usually not more than 5% of the net radiation while over land surfaces ΔF is negligible and net heat storage term G , may be neglected or regarded as constant over a long period.

Make a gain

LATENT HEAT Flux

Latent heat flux is the flux of heat from the earth surface to the atmosphere that is associated with evaporation of water & surface and subsequent condensation of

Vapour in the atmosphere it is an important component of earth's surface energy budget. latent heat flux is commonly measured with the ~~Bowen~~ Bowen ratio technique or by eddy covariance.

RADIATION BUDGET

Radiation balance is the difference between

solar radiation and effective outgoing radiation. It adds up to zero for the earth's atmosphere system over a period of a year. This prevents the earth's from getting warmer or cooler. There is an excess of radiation balance in the low latitudes and marked deficits in the high latitudes.

On the other hand, radiation balance in the atmosphere between the amount of radiation which are absorbed and emitted by a given surface body. It is also known as the net radiation and can be expressed in term of radiation balance equation as follows for the earth's surface.

$$R_n = (Q + q)(1 - \alpha) + [I_n - I_r]$$

where

R_n = radiation balance or net radiation

$(Q + q)$ = Insolation consisting of the direct horizontal and diffuse radiation

α = albedo of the earth's surface

IV = Counter radiation from the atmosphere

IT = Terrestrial radiation

The atmosphere is a radiative sink in

All latitudes where the earth's surface is a

Heat source in all latitudes except the polar

zone. To prevent the earth's surface from warming

and the atmosphere from cooling, surplus energy

is continually transferred from the earth's surface

to the atmosphere.

Vertical exchange - This vertical exchange

of heat takes place in the following major ways

1. Evaporation of water from the earth's surface and the atmosphere is released the latent heat.

ii. Conduction of sensible heat from the earth's surface to the atmosphere.

iii. Convection involving turbulent diffusion of heat into the atmosphere from the earth's surface.

Polewards of about 40 latitude, the radiative

balance of the earth's atmosphere system is -ve while it is +ve between 40° latitude north

and south of the equator. To prevent the

the low latitude from getting warmer and the polar zones from colder, there is a horizontal

Transfer of surplus energy from the low to the middle and high latitudes. This horizontal exchange of energy is also partly induced by the differential heating of continents and oceans. It is effected mainly by:
 1. The poleward transport of sensible heat by atmospheric circulation and warm ocean currents from the low latitudes.
 2. The release of latent heat when water vapour carried poleward from the low latitudes condenses in the atmosphere.
 Thus, there is a two-way heat transfer within the earth-atmosphere system, from the earth's surface to the atmosphere and from the equator to the poles. This heat transfer is the physical basis from the ~~the~~ existence of the general circulation of atmosphere and ocean water.

1. Definition or short or explanation of the following.
 - i. Meteorology, Atmosphere, Meteorological station, climate.
 - ii. Diurnal variation, Atmospheric temperature, wind velocity, humidity, specific, relative humidity and their periods.
 - iii. Barzale, sleet, snow, solar radiation, temperature gradient, lapse rate, Atmospheric pressure, Micro climate, solar energy.

2. Explanation of the following.

- i. Energy budget ~~con~~ concept ii. Latent heat flux
- iii. Radiation balance iv. Two-wings heat transfer.
- iv. Net radiation

3. Vertical exchange and horizontal exchange of energy.
4. Classification of crops on their basis of photoperiodism.
5. Examples of light forming plants & shade loving plants.
6. Photoperiod and photoperiodism.

7. The concept growing degree-day and its assumption.

8. Importance of moisture in agriculture and soil moisture control.

9. Weather phenomena that constitute hazard to agriculture.

10. Influence of temperature on crops.

11. Use and value of agroclimatology studies in agriculture and development in farming.

12. climatic consequences of global warming.

13. about radiation and crops relation at up.

14. Importance of solar energy.

15. Right values for optimum growth and fruiting of some crops.

16. Most appropriate site for meteorological station.

17. Instruments used in measuring insulation & their use.

18. Effect of climate & their relation.

19. Value of climatic elements.

20. Melting and boiling point of water is 32°F & 100°C .

21. Terms of solar radiation.

Examine the influence of weather on the incidence of crop pests and diseases, especially the head mould of sorghum, Rosette and other diseases that attack grain. mildew rust, scab and blight, corn ear yellow mosaic disease, black pod disease